Roll No. | | | | | | | |

ODD END TERM EXAMINATION, December - 2024

Name of the Course: **B.Tech., ME** Semester: **VIIth**

Name of the Paper: **Total Quality Management** Paper Code: **(DEEM 06)**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	**(20 marks)**	
(a)	Compare and contrast the philosophies of Deming, Juran, and Crosby in the context of quality management. Discuss how their approaches can collectively contribute to improving organizational performance.	**CO1, CO3, and CO4**
(b)	Explain the significance of the PDCA and PDSA cycles in continuous improvement. Illustrate their application in a real-world scenario or case study where these cycles helped enhance service quality and customer satisfaction.	
(c)	Discuss the role of employee involvement in quality management with reference to motivation, empowerment, and performance appraisal. How does this impact customer retention and the overall perception of service quality? Provide examples or a case study to support your answer.	
Q2	**(20 marks)**	
(a)	Discuss the dimensions of quality with examples, and explain how they contribute to achieving customer satisfaction.	**CO1, CO2, and CO3**
(b)	What are the principles of Total Quality Management (TQM)? Illustrate its benefits and scope with a relevant case study.	
(c)	Explain the concept of quality costs, and discuss the barriers to TQM implementation with suitable examples.	
Q3	**(20 marks)**	
(a)	i) Explain the Seven Basic Tools of Quality and discuss how each tool can be utilized to identify and solve quality-related issues in a production process. ii) Provide an example of how software like Minitab can enhance the effectiveness of at least two of these quality tools.	**CO1, CO2, and CO5**
(b)	i) Describe the concepts of Measures of Central Tendency and Dispersion in statistical analysis. How are they essential in understanding a population versus a sample in quality control? (5 Marks) ii) Explain the significance of the Normal Curve in quality management and	

	how it relates to Process Capability and Six Sigma methodologies. (5 Marks)	
(c)	i) Define Reliability in the context of quality management. Illustrate the Bathtub Curve, labelling each phase, and explain the failure rates associated with each phase. ii) Derive the hazard function for an exponential probability distribution and discuss its application in determining system reliability for components arranged in series and parallel configurations.	
Q4	**(20 marks)**	
(a)	Discuss the need for ISO 9000 in modern industries and explain how the implementation of ISO 9000:2000 Quality System helps in achieving organizational goals. Include examples to support your answer.	**CO4,**
(b)	Explain the key elements of ISO 9000:2000 Quality System and describe the steps involved in implementing this system in an organization. Highlight the importance of documentation and quality auditing in the process.	**CO3, and CO6**
(c)	Compare ISO 14000 and TS 16949 standards in terms of their concept, requirements, and benefits. Provide a case study or real-life example where one of these standards was successfully implemented.	
Q5	**(20 marks)**	
(a)	Explain the Benchmarking Process in detail and discuss its significance by providing examples of how organizations use benchmarking to improve their processes and achieve competitive advantages.	**CO2, CO4,**
(b)	With the help of a diagram, explain the concept of the House of Quality in Quality Function Deployment (QFD). Discuss the steps in the QFD process and outline its benefits in product and service design.	**and CO3**
(c)	Compare breakdown maintenance and preventive maintenance in terms of their advantages and disadvantages. Describe the principles of Total Productive Maintenance (TPM) and how it enhances overall equipment effectiveness in organizations.	